

INFORMATION ARCHITECTURE / PRODUCT DESIGN BASELINE

AL-AZHARI LIBRARY

Information Architecture & Sitemap

Complete Website, Customer Portal & Admin Architecture

مكتبة الأزهرى

Purpose	Production-ready translation of the approved business and functional requirements into navigable information architecture.
Source basis	Master Product Brief v2.0 + Product Requirements Document v1.0.
Functional authority	PRD v1.0 for finalized behavior; Master Product Brief for business context where the PRD is silent.
Status	Architecture baseline for stakeholder review; unresolved items are explicitly marked [OPEN DECISION] .
Prepared	22 September 2026

This document does not introduce reviews, ratings, an LMS, marketplace behavior, automated payments, Google Maps address entry, phone OTP, or in-site service file uploads.

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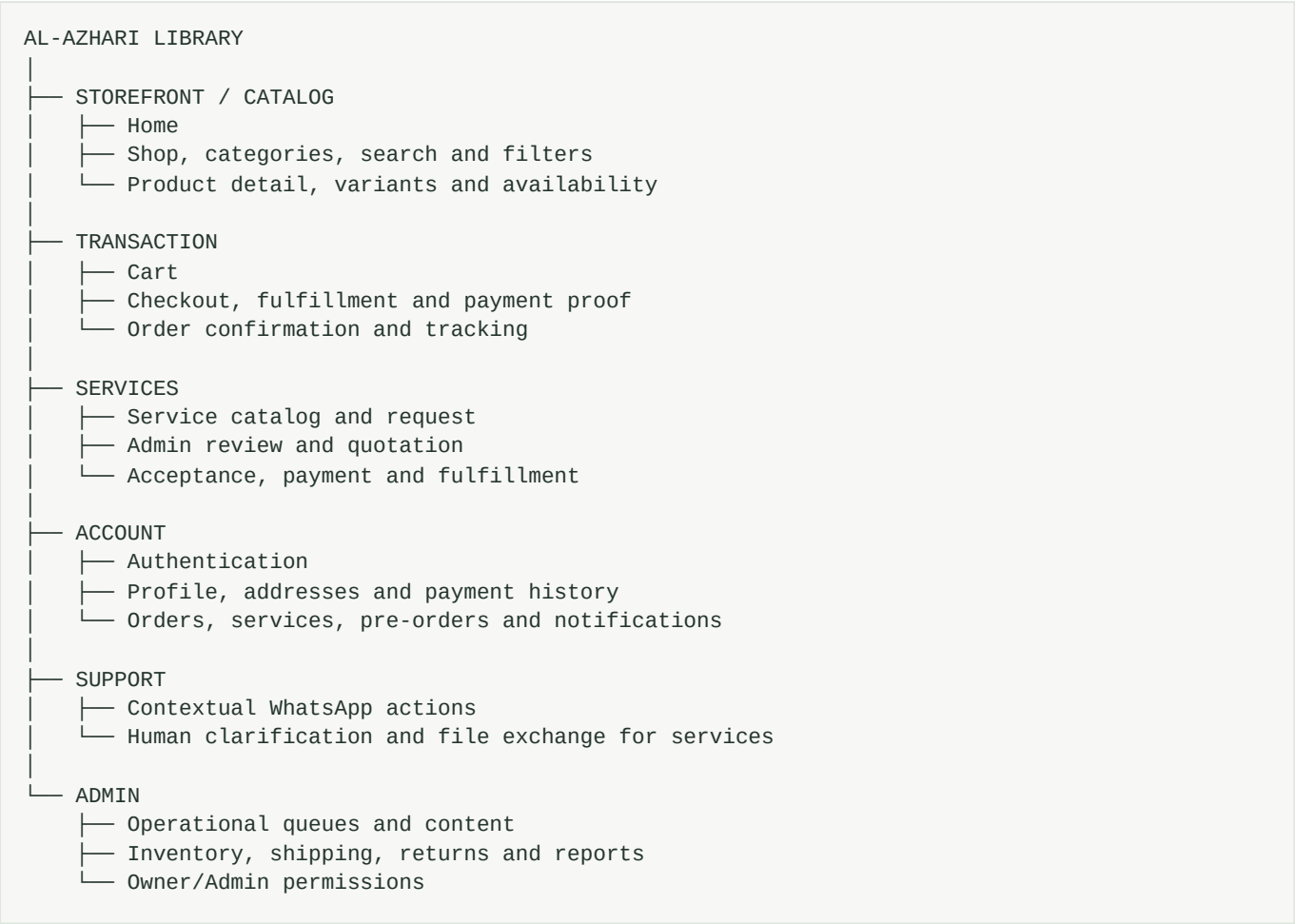
[OPEN DECISION]

SOURCE STATUS

The supplied PRD is labeled Draft — for stakeholder review. This IA preserves its functional decisions while keeping unresolved policy and operational questions visible.

1. Architecture overview

The platform is one digital extension of the physical library, organized into six product areas. The areas are connected by customer journeys, but they are not interchangeable: fixed-price products use cart and checkout; variable-price services use request and quotation; Admin operates the system of record behind both.



What each area is for

Area	Customer or business need	Connected to	Must remain separate from
Storefront / Catalog	Discover the full library, especially books; answer what it is, what it costs and whether it is available.	Transaction, Services, Support, SEO/social entry points.	Seasonal campaigns must not replace the permanent taxonomy; Services do not become ordinary products.
Transaction	Complete a product order with transparent fulfillment and manual payment verification.	Catalog → Cart → Checkout → Order → Tracking → Account.	Service quotation lifecycle; WhatsApp is not the transaction record.
Services	Request printing, photocopying, binding, applications, transfers and research/formatting help when price depends on the case.	Service request → Admin review → Quotation → Payment → Fulfillment.	Fixed-price product/cart/checkout model; no in-site service file uploads.
Account	Give registered customers durable visibility into profile, orders, payment history, service requests and notifications.	Orders, services, pre-orders, returns and notifications.	Guest checkout must not be blocked by account creation.
Support	Preserve the trusted human relationship for clarification, edge cases and service file exchange.	Persistent/contextual WhatsApp actions across customer and Admin views.	WhatsApp must not replace authoritative order/service records or automated notification channels.

Area	Customer or business need	Connected to	Must remain separate from
Admin	Operate products, orders, payments, inventory, shipping, services, returns, content, reports and permissions.	All domain areas; database remains source of truth.	Admin is isolated from public navigation and gated by RBAC.

Primary navigation hierarchy

Global navigation: Home, Shop/Catalog, Services, Search, Account, Cart and persistent WhatsApp access. Books lead the storefront order; supplies and other products remain discoverable without competing with the core.

Contextual navigation: category breadcrumbs, product-to-cart/pre-order actions, checkout step navigation, order timeline actions, service request/quotation actions, and return access from completed orders.

Account navigation: Dashboard, Profile, Orders, Addresses, Payment History, Service Requests, Pre-orders and Notifications.

Admin navigation: a separate authenticated shell with operational queues first, then catalog, customers, services, inventory/shipping, returns/refunds, content, reports and settings/RBAC.

CONFIRMED BOUNDARY	Services are structurally separate from the normal product/cart/checkout flow because final pricing is determined after the library reviews the request. Seasonal promotions are a presentation layer over stable navigation and taxonomy.
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2. Structural principles and domain boundaries

Books-first, not books-only

Books are the core business and default merchandising priority. The active MVP taxonomy is Books; School/Study Products; Other Products — games, gifts and gift wrapping; and Services as a separate product type. Azhar specialization is the anchor, while the audience remains open to general-education students, university students and parents.

Stable catalog, seasonal presentation

July summaries, August applications/transfers and September school supplies may influence homepage modules, featured products, banners and category highlighting. They must not reorder the permanent site into a season-only store or create temporary top-level taxonomy.

System-of-record rule

Website order and service-request records are authoritative. WhatsApp is a support and communication channel. Admin actions that change payment, order, refund or fulfillment state remain visible in the relevant record and retain an audit trail where required.

Domain model at a glance

Product commerce

- Product / variant
- Cart
- Order
- Payment / proof
- Inventory reservation
- Fulfillment / tracking
- Return / refund

Student services

- Service category
- Service request
- Admin review
- Quotation
- Customer accept/reject
- Payment / proof if applicable
- Processing / completed

Cross-cutting

- Customer account
- In-app notifications
- Email verification/reset/transactional messages
- WhatsApp actions
- RBAC
- Reports

Excluded from this IA

- Reviews and ratings
- Automated payment gateway
- Google Maps address pin
- Phone OTP
- LMS/course features
- Marketplace/multi-seller features
- In-site service file upload
- Dedicated search engine or microservices

Source hierarchy used in this document



A recommendation is not silently promoted to a requirement. For example, re-validating a cart at checkout, requiring variant selection before add-to-cart, showing a full order breakdown, auto-closing rejected quotations and preventing negative stock are identified as recommendations in the PRD and should receive owner approval before implementation as binding behavior.

3. Complete sitemap

This is the route-level architecture. The master inventory below is the implementation index; detailed behavior and state rules are described in the following sections.

/	/
├ Home	/shop
├ Shop / Catalog	/shop/all
│ └ All Products	/categories
│ └ Categories	/categories/:slug
│ │ └ Category Listing	/search?q=...
│ └ Search Results	/products/:slug
│ └ Product Detail	/products/:slug/pre-order
│ │ └ Pre-order Request	/cart
├ Cart	/checkout/*
├ Checkout	/checkout/customer
│ └ Customer Information	/checkout/fulfillment
│ └ Address & Fulfillment	/checkout/payment
│ └ Payment	/checkout/payment-proof
│ └ Payment Proof	/checkout/review
│ └ Order Review	/orders/:id
├ Orders	/orders/:id/confirmation
│ └ Confirmation	/orders/:id/tracking
│ └ Tracking	/orders/:id/return
│ └ Return Request	/services
├ Services	/services/catalog
│ └ Service Catalog	/services/:slug
│ └ Service Detail	/services/:slug/request
│ └ Service Request	/account
├ Account	/account/profile
│ └ Profile	/account/orders
│ └ Orders	/account/addresses
│ └ Addresses	/account/payments
│ └ Payment History	/account/services
│ └ Service Requests	/account/pre-orders
│ └ Pre-orders	/account/notifications
│ └ Notifications	/login, /register, /verify-email, /forgot-password
├ Authentication	/contact
├ Support	/admin/*
├ Admin	
│ └ Dashboard, Orders, Products, Categories, Customers, Payments	
│ └ Services, Pre-orders, Inventory, Shipping, Returns, Refunds	
│ └ Notifications, Content, Coupons, Reports, Settings/RBAC	
│ └ Unauthorized / Forbidden / Not Found states	

Master page inventory

Auth is shown as the minimum required to enter the page. “Conditional” means a guest may see the page through a reference/confirmation path, while account-only history and action access remains protected.

ID	Area	Page / route	User	Auth	Public	Flow	Purpose / primary actions	Related	
PUB-001	Public	Home	/	Guest; Customer	No	Yes	No	Global entry point; brand, books-first discovery, seasonal modules, services preview, WhatsApp	Shop; Services; Search; Contact
PUB-002	Public	Shop / Catalog	/shop	Guest; Customer	No	Yes	No	Permanent catalog entry; books lead default merchandising	Category; Search; Product

ID	Area	Page / route	User	Auth	Public	Flow	Purpose / primary actions	Related	
CAT-001	Catalog	All Products	/shop/all	Guest; Customer	No	Yes	No	Browse all active purchasable products	Product; Cart
CAT-002	Catalog	Category Listing	/categories/:slug	Guest; Customer	No	Yes	No	Browse a stable category without knowing exact terminology	Search; Product
CAT-003	Catalog	Search Results	/search?q=...	Guest; Customer	No	Yes	No	Search by name, author, grade, stage, subject, publisher, ISBN, category	Filters; Product
CAT-004	Catalog	Filtered / Sorted Results	/search?...	Guest; Customer	No	Yes	No	Apply education type, stage, grade, subject, publisher, price, availability	Product
CAT-005	Catalog	Product Detail	/products/:slug	Guest; Customer	No	Yes	Conditional	Evaluate identity, price, availability, variants and purchase/pre-order action	Cart; Pre-order; WhatsApp
TRX-001	Transaction	Cart	/cart	Guest; Customer	No	Yes	Yes	Review multiple products/variants, quantities, discount and proceed	Checkout; Shop
TRX-002	Transaction	Checkout — Customer Information	/checkout/customer	Guest; Customer	No	No	Yes	Collect recipient and contact details; guest checkout remains available	Address; Cart
TRX-003	Transaction	Checkout — Address & Fulfillment	/checkout/fulfillment	Guest; Customer	No	No	Yes	Collect detailed address or select free pickup; show delivery estimate	Payment; Order review
TRX-004	Transaction	Checkout — Payment	/checkout/payment	Guest; Customer	No	No	Yes	Select supported method; display proof step only where applicable	Order review
TRX-005	Transaction	Payment Proof	/checkout/payment-proof	Guest; Customer	No	No	Yes	Upload one or more payment screenshots for InstaPay/e-wallets; no transaction ID	Payment; Review
TRX-006	Transaction	Order Review	/checkout/review	Guest; Customer	No	No	Yes	Show items, product total, shipping estimate, discounts and final total before submit	Submit; Checkout
TRX-007	Transaction	Order Confirmation	/orders/:id/confirmation	Guest; Customer	No	Conditional	Yes	Confirm receipt, reference, initial Pending Review state and next step	Order detail; WhatsApp
ORD-001	Orders	Order Detail	/orders/:id	Guest; Customer	Conditional	Conditional	Yes	Single source of truth for items, status, payment, fulfillment and costs	Tracking; Return; WhatsApp
ORD-002	Orders	Order Tracking	/orders/:id/tracking	Guest; Customer	Conditional	Conditional	Yes	State timeline from submission through delivery/pickup/completion	Order detail; Notifications
SRV-001	Services	Services Landing	/services	Guest; Customer	No	Yes	No	Separate entry to variable-price student services	Service catalog; WhatsApp
SRV-002	Services	Service Catalog	/services/catalog	Guest; Customer	No	Yes	No	Browse confirmed service categories	Service detail; Request
SRV-003	Services	Service Detail	/services/:slug	Guest; Customer	No	Yes	No	Explain request-based lifecycle; no fixed final price before quotation	Request; WhatsApp/Telegram
SRV-004	Services	Service Request	/services/:slug/request	Guest; Customer	No	Conditional	No	Describe need; fields vary by service and remain partly open	Request detail
SRV-005	Services	Service Request Detail	/account/services/:id	Customer	Yes	No	Separate lifecycle	View status, quotation, communication and payment when available	Quote; Notifications
SRV-006	Services	Quotation Review	/account/services/:id/quotation	Customer	Yes	No	Separate lifecycle	View final quoted price and Accept/Reject	Payment; Service detail
PRE-001	Pre-order	Pre-order Request	/products/:slug/pre-order	Guest; Customer	Conditional	Conditional	Yes	Request a product/variant that is out of stock and marked eligible	Product; Account

ID	Area	Page / route	User	Auth	Public	Flow	Purpose / primary actions	Related	
PRE-002	Pre-order	Pre-order Detail	/account/pre-orders/:id	Customer	Yes	No	Yes	View review, payment, expected availability if entered and fulfillment status	Notifications; Order
ACC-001	Account	Login	/login	Guest	No	No	No	Login using email or phone plus password	Dashboard; Forgot password
ACC-002	Account	Register	/register	Guest	No	No	No	Create account with name, phone, email and password	Verification; Dashboard
ACC-003	Account	Email Verification	/verify-email	Customer	Conditional	No	No	Complete email verification; exact gating scope is open	Login; Dashboard
ACC-004	Account	Forgot Password	/forgot-password	Guest	No	No	No	Request password reset by email	Reset password
ACC-005	Account	Reset Password	/reset-password	Guest	Conditional	No	No	Set a new password from email flow	Login
ACC-006	Account	Dashboard Overview	/account	Customer	Yes	No	No	Entry to account areas and current activity	Orders; Services; Notifications
ACC-007	Account	Profile	/account/profile	Customer	Yes	No	No	View/edit profile information	Dashboard
ACC-008	Account	Orders	/account/orders	Customer	Yes	No	No	List order history and current states	Order detail; Tracking
ACC-009	Account	Addresses	/account/addresses	Customer	Yes	No	No	Manage one or more saved addresses	Checkout
ACC-010	Account	Payment History	/account/payments	Customer	Yes	No	No	View payment history connected to account orders	Order detail; Refund
ACC-011	Account	Service Requests	/account/services	Customer	Yes	No	Separate lifecycle	List service requests and quotation/payment states	Service detail
ACC-012	Account	Pre-orders	/account/pre-orders	Customer	Yes	No	Yes	List pre-order requests and progress	Product; Order
ACC-013	Account	Notifications	/account/notifications	Customer	Yes	No	No	List in-app lifecycle notifications	Order; Service; Payment
RET-001	Returns	Return Request	/orders/:id/return	Customer	Yes	No	Yes	Select eligible item(s), reason and evidence where appropriate	Return detail
RET-002	Returns	Return Detail / Refund Record	/account/returns/:id	Customer	Yes	No	Yes	View return state and recorded manual refund	Order; Payment history
SUP-001	Support	Contact / WhatsApp	/contact	Guest; Customer	No	Yes	No	General contact access; phone number remains open	WhatsApp
ADM-001	Admin	Admin Dashboard	/admin	Admin; Owner	Yes	No	No	Operational overview and priority queues	All admin modules
ADM-002	Admin	Orders Queue	/admin/orders	Admin; Owner	Yes	No	Yes	Pending Review, payment and fulfillment queues	Order detail
ADM-003	Admin	Order Detail	/admin/orders/:id	Admin; Owner	Yes	No	Yes	Review, accept/reject, verify payment, adjust shipping, advance status	Customer; WhatsApp
ADM-004	Admin	Products	/admin/products	Admin; Owner	Yes	No	No	List/search/manage products and publication	Product edit; Inventory
ADM-005	Admin	Product Editor / Variants	/admin/products/:id/edit	Admin; Owner	Yes	No	No	Edit identity, price, images, description, availability and variants	Inventory
ADM-006	Admin	Categories	/admin/categories	Admin; Owner	Yes	No	No	Manage stable catalog categories; future expansion remains inactive in MVP	Products
ADM-007	Admin	Customers	/admin/customers	Admin; Owner	Yes	No	No	View customer profiles and associated operational records	Customer detail
ADM-008	Admin	Payments Queue	/admin/payments	Admin; Owner	Yes	No	Yes	Inspect proof screenshots; Confirm/Reject/Request New Proof	Order detail

ID	Area	Page / route	User	Auth	Public	Flow	Purpose / primary actions	Related	
ADM-009	Admin	Services	/admin/services	Admin; Owner	Yes	No	Separate lifecycle	Manage service catalog and request queue	Request; Quote
ADM-010	Admin	Service Request Detail	/admin/services/requests/:id	Admin; Owner	Yes	No	Separate lifecycle	Review, clarify via WhatsApp, quote, update state	Customer; WhatsApp
ADM-011	Admin	Pre-orders	/admin/pre-orders	Admin; Owner	Yes	No	Yes	Review/accept/reject requests; mark availability to trigger fulfillment	Product; Order
ADM-012	Admin	Inventory	/admin/inventory	Admin; Owner	Yes	No	No	View stock, reserved vs available; adjust stock	Products; Orders
ADM-013	Admin	Shipping	/admin/shipping	Admin; Owner	Yes	No	Yes	Configure/operate provider selection and cost review; rate source remains open	Order detail
ADM-014	Admin	Returns	/admin/returns	Admin; Owner	Yes	No	Yes	Review evidence and approve/reject return requests	Refund record
ADM-015	Admin	Refund Record	/admin/refunds/:id	Admin; Owner	Yes	No	Yes	Record manual amount, method and timestamp	Return; Payment
ADM-016	Admin	Notifications	/admin/notifications	Admin; Owner	Yes	No	No	View operational notifications and event activity	Orders; Services
ADM-017	Admin	Content / Merchandising	/admin/content	Admin; Owner	Yes	No	No	Edit descriptions and seasonal/homepage modules without changing taxonomy	Home; Products
ADM-018	Admin	Coupons & Discounts	/admin/coupons	Admin; Owner	Yes	No	No	Manage percentage/fixed, seasonal, product/category discounts; unresolved governance configurable later	Checkout; Reports
ADM-019	Admin	Reports	/admin/reports	Admin; Owner	Yes	No	No	View orders, revenue, geography, payment methods, services, demand and coupons	Dashboard
ADM-020	Admin	Settings / RBAC	/admin/settings	Owner; Admin	Yes	No	No	Operational settings and permission boundary; owner-only actions only when defined	RBAC
ERR-001	System	Unauthorized / Forbidden	/403	Any	No	No	No	Explain denied access and provide safe navigation back	Login; Home
ERR-002	System	Not Found	/404	Any	No	No	No	Handle unknown route/product/order without exposing data	Home; Search

4. Public storefront and catalog

Home — /

The home page is the public front door and the social/referral landing point. It should answer “what is this library, what can I get here, and where do I start?” without becoming a generic content portal.

Order	Module	Purpose / IA rule
1	Header	Logo, Shop, Services, Search, Account, Cart and WhatsApp. Keep the primary route to books visible.
2	Hero / primary discovery	Communicate the library and online ordering; primary action to Shop/Catalog. Copy must not make unverified “best/number one” claims.
3	Books-first discovery	Featured or highlighted book products/categories; default merchandising priority is Books.
4	Stable category access	Books; School/Study Products; Other Products. Services gets a distinct route and treatment.
5	Seasonal presentation	Temporary banners/featured categories/products for summaries, applications/transfers or school supplies; never replace full catalog access.
6	Services preview	Printing, photocopying, binding and student administrative/research-formatting services; explain that a request and quotation precede final price.
7	Trust and practical information	Professional, calm presentation; availability and price clarity; pickup and cash/support options where confirmed.
8	WhatsApp CTA	Persistent/contextual contact action; support channel, not order record.
9	Footer	Contact/location and business information that is confirmed; final phone numbers remain open.

Catalog and category architecture

The catalog must support discovery by browsing or by search. The permanent structure is intentionally small enough for students and parents to understand, while the data model can support future categories without redesign. Future Tajweed, Qira'at, Qur'an memorization, nursery curricula and Qurans are not active or purchasable in MVP.

Catalog surface	Required content / behavior	Primary actions
Shop / all products	Active product cards with name, image, price, availability and applicable variant context.	Open product; filter; sort if defined by UI; add eligible product to cart.
Category listing	Stable category context; Books visually first; product cards and search/filter access.	Browse; refine; open product.
Search	Search product name, author, grade, stage, subject, publisher, ISBN and category. Missing fields must not make a product undiscoverable by populated fields.	Submit query; clear query; open result.
Filters	Education type, stage, grade, subject, publisher, price range, availability.	Apply/remove filters; retain visible active filters.
Availability	In Stock; Out of Stock; Pre-order-eligible. State must be unambiguous.	Add to cart; pre-order; understand unavailable state.
Sorting	A sort control may be used only for supported catalog ordering; no unconfirmed ranking logic is prescribed by the sources.	Choose available sort; preserve filters.

[OPEN DECISION]
CATALOG POLICY

The source documents do not define a full sort order, exact product field completeness, or additional taxonomy beyond the confirmed categories. Do not invent customer-facing filters or ranking rules during wireframing.

5. Product details and states

Product detail — /products/:slug

The product page converts the most repetitive staff questions into self-service: what is it, what does it cost, is it available, and can the customer order it now?

Information group	Content	Interaction / rule
Identity	Product name, category, image(s), concise description where available.	Description may be absent in MVP if Admin has not authored it; do not block publication solely on description.
Educational metadata	Applicable author, grade, stage, subject, publisher, ISBN and education type fields.	Show populated fields; missing fields do not prevent discovery.
Commercial	Price and applicable variant price.	Price can change; variant price is independent where variants exist.
Availability	In Stock, Out of Stock, or Pre-order-eligible.	In Stock → Add to Cart. Eligible out of stock → Pre-order. Out of stock and not eligible → clear unavailable message without purchase action.
Variants	Publisher, edition, part, language or other commercially meaningful attributes where applicable.	Show applicable variants; do not assume every product has every attribute. Variant-specific price and availability.
Support	Contextual WhatsApp action.	Support only; does not create the authoritative order.
Related navigation	Breadcrumb back to category/search; cart or pre-order destination.	No reviews or ratings.

Required product states

State	Visible information	Primary action
Loading	Skeleton or loading indicator for identity, price and availability.	Wait; preserve route context.
In Stock	Identity, price, current availability, applicable variant state.	Select required variant where applicable; Add to Cart.
Out of Stock / pre-order eligible	Unavailable state plus pre-order explanation; no Add to Cart.	Pre-order Request.
Out of Stock / unavailable	Clear unavailable state; no purchase CTA.	WhatsApp support where appropriate; continue browsing.
Variant selected	Selected attribute(s), price and availability update.	Add to Cart or Pre-order based on selected variant.
Product not found	Not-found explanation without exposing internal data.	Return to Shop/Search/Home.
Error	Recoverable loading/data failure message.	Retry; fallback to Shop/WhatsApp.

EXCLUDED

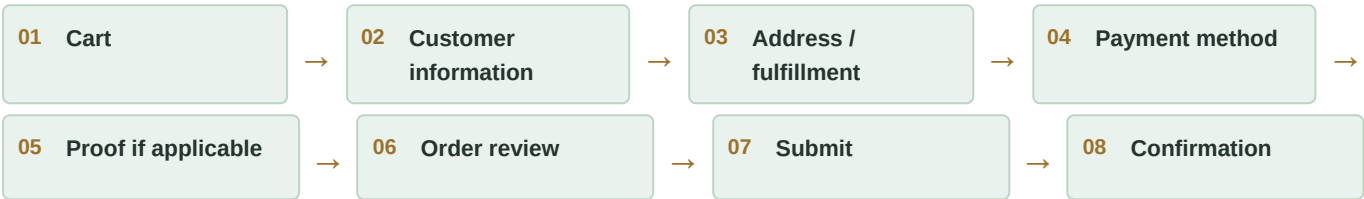
Product reviews and ratings are explicitly outside the current scope and must not appear as a placeholder section.

6. Cart, checkout, payments and orders

Cart — /cart

The cart supports multiple distinct products/variants and has no minimum order value. It is the bridge from fixed-price catalog discovery to transactional checkout. Before checkout, the UI should surface quantity, line price, availability, discount/coupon result and totals. Re-validation at checkout is a PRD recommendation, not yet a settled requirement.

Checkout sequence



Step	Required architecture	State / rule
Customer information	Guest or registered checkout; recipient name and phone; customer details.	Guest checkout is never blocked by account creation.
Address	Governorate, city, area, street, building number, floor, apartment, landmark, recipient name, phone and notes.	No Google Maps or location pin. Registered customers may reuse saved addresses.
Fulfillment	Delivery or Pickup.	Pickup is free. Delivery shows estimated shipping cost by governorate/location before confirmation.
Payment	PRD functional list: InstaPay, Vodafone Cash, Orange Cash, Etisalat Cash, WE Pay and Cash on Delivery.	Final list has a source discrepancy; see OD-20 below. No automated gateway.
Payment proof	For InstaPay/e-wallet: display library payment details; customer transfers outside platform; upload one or more screenshots.	No transaction ID/reference number. Proof is reviewed manually after Admin accepts the order.
Order review	Items, quantities, product total, shipping cost, discounts and final total.	Full breakdown is a product recommendation but is the safe review architecture.
Submit	Create order and reference; initial state is Pending Review.	Customer and Admin receive notification; state is authoritative in database.

[OPEN DECISION] PAYMENT-METHOD DISCREPANCY

The Master Product Brief confirms COD, cash at library, InstaPay and Vodafone Cash. The PRD's explicit list uses COD plus four e-wallets and omits cash at library. This IA follows the PRD list as functional authority, but the owner must confirm the final list before implementation.

Order lifecycle and tracking

Pending Review

- └ Admin accepts → Accepted
 - └ digital payment → Awaiting Payment → Payment Verification
 - └ Confirmed → Payment Confirmed → Preparing
 - └ Rejected → Awaiting New Proof → re-upload → Payment Verification
 - └ COD → Customer Confirmation Required → customer confirms → Confirmed → Preparing
- └ Admin rejects → Rejected [terminal]

Preparing

- └ Pickup → Ready for Pickup → Picked Up → Completed
- └ Delivery → Shipped → Out for Delivery → Delivered → Completed

Pending Review → Customer cancellation → Cancelled [terminal]

Completed → Return Request → Returned / Refund lifecycle

Order state	Customer-facing page behavior	Allowed action / next state
Pending Review	Show submitted items, costs and review status.	Edit/cancel only within permitted window; Admin accept/reject.
Accepted	Show accepted state and next payment/COD confirmation step.	No customer cancellation after acceptance.
Awaiting Payment / Payment Verification	Show payment instructions, proof status and upload/re-upload when applicable.	Upload proof; Admin Confirm/Reject/Request New Proof.
Awaiting New Proof	Explain rejection/request and provide new screenshot path.	Re-upload → Payment Verification.
Customer Confirmation Required	Prompt COD customer to confirm after Admin acceptance.	Confirm → Confirmed. Timeout behavior is open.
Preparing	Show preparation progress.	Admin advances to pickup or shipping path.
Ready for Pickup / Picked Up	Show pickup is ready, then completed collection.	Admin marks collection.
Shipped / Out for Delivery / Delivered	Show delivery progress without inventing delivery promises.	Admin/system advances state.
Completed	Close order and expose eligible return access.	Request Return if product/category permits.
Rejected / Cancelled	Terminal state with reason where captured.	No further order progression.

Shipping and pickup

The customer selects Delivery or free Pickup. The library, not the customer, selects the provider for delivery (for example Bosta, Egyptian Post or a local partner). The system shows an estimate at checkout; Admin can adjust the final cost after submission. Exact rate source, binding point, delivery times and excluded areas are [OPEN DECISION].

7. Authentication and customer account

Authentication routes

Route	Purpose	Rules
/register	Create registered customer account.	Name, phone, unique valid email and password. No phone OTP.
/verify-email	Verify account email.	Verification is required; exact gating scope for full privileges vs registration completion is [OPEN DECISION].
/login	Start session.	Email + password or phone + password.
/forgot-password → /reset-password	Recover account.	Email-based reset.
Guest checkout	Buy without registration.	Do not force account creation; confirmation and order reference still need a usable path.

Customer dashboard — /account

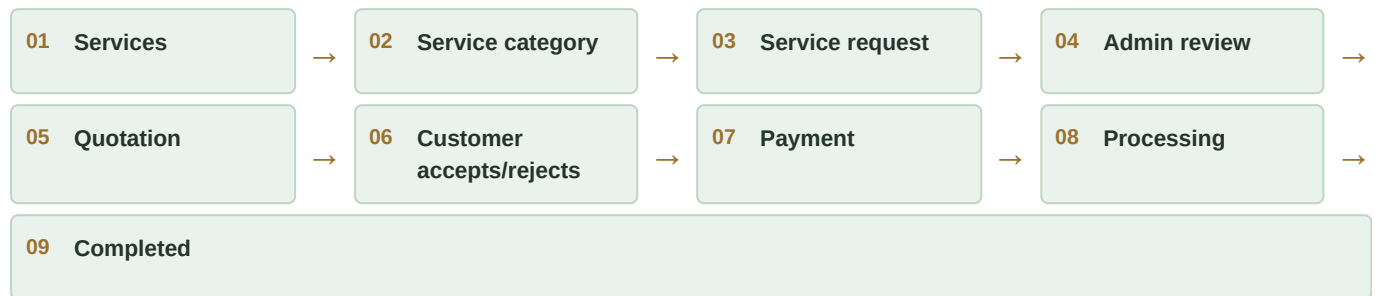
The dashboard is the durable relationship layer for registered customers. It should use a simple, mobile-friendly section hierarchy and show current operational state before secondary account settings.

Section	Route	Primary content / actions
Overview	/account	Current orders, service requests, notifications and shortcuts.
Profile	/account/profile	View/edit profile information.
Orders	/account/orders	Order history by state; open detail/tracking; cancel only Pending Review; edit where permitted.
Addresses	/account/addresses	Create/edit/delete one or more saved addresses; no map pin.
Payment History	/account/payments	Payment records linked to orders; distinguish original payment, proof and refund record.
Service Requests	/account/services	Request list, detail, quotation, accept/reject, payment and processing state.
Pre-orders	/account/pre-orders	Request, review, payment, availability and fulfillment progress.
Notifications	/account/notifications	In-app lifecycle notifications; real-time delivery is a convenience, database is truth.

ACCOUNT BOUNDARY	The customer dashboard exposes a customer's own records. Payment-proof screenshots and personal data are restricted to authorized Admin/Owner access on the operational side.
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8. Student services architecture

Services are not a product category hidden inside Shop. They are a separate product type and a separate customer mental model because the library cannot promise a final price until it understands the case.



Confirmed service catalog

Service category	Entry / explanation	What remains open
Printing	Request-based service; customer describes need.	Required fields, file expectations, pricing model and turnaround.
Photocopying	Request-based service; no fixed final price before review.	Required fields, file expectations, pricing model and turnaround.
Binding	Request-based service; no fixed final price before review.	Required fields, file expectations, pricing model and turnaround.
Applications / transfers	Library performs the administrative service on behalf of the customer; not an end-to-end self-service form.	Required information/documents, pricing, turnaround and any additional rules.
Research / formatting help	Request-based help reviewed by the library.	Exact scope, required information, pricing and turnaround.
Other administrative services	Only when supported by the confirmed service scope and owner decision.	Any future service beyond the confirmed list is [OPEN DECISION].

Service request pages

Page	Purpose	Must show
/services	Explain service area and why it differs from product shopping.	Confirmed categories, request CTA, WhatsApp/Telegram file-exchange explanation.
/services/:slug	Describe the chosen service.	Request-based lifecycle; do not show a final fixed price before quotation.
/services/:slug/request	Collect a structured description.	Only fields approved for that service. No file-upload control.
/account/services/:id	Track request.	Submitted, Admin Review, Quotation Sent, Awaiting Payment, Payment Verification, Payment Confirmed, Processing, Completed or terminal closed state.
/account/services/:id/quotation	Make the decision visible.	Final quoted price; Accept or Reject; payment only after acceptance.

Files and communication

No service-related file is uploaded to or stored by the platform. The request page must direct the customer to exchange files through WhatsApp or Telegram when needed. Admin may use WhatsApp to clarify the request, but the service request record remains the source of truth.

**[OPEN DECISION]
SERVICE FORM
DESIGN**

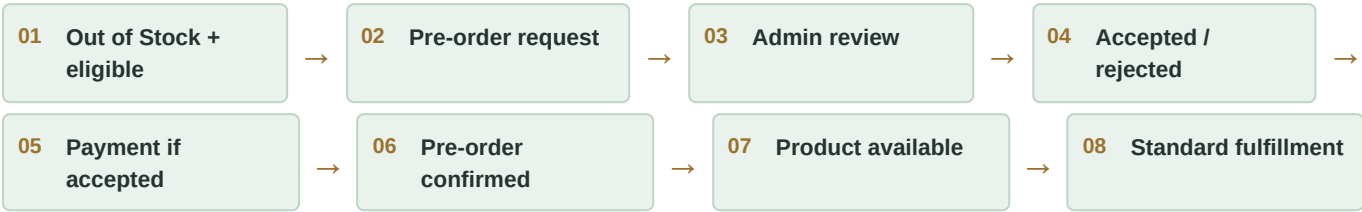
The exact information/documents required per service type, file-handling expectations and turnaround times are not defined. The IA reserves service-specific form space without inventing fields.

Quotation behavior

Admin reviews the request and sends a quotation. The customer can Accept or Reject. On acceptance, payment becomes available. On rejection, the request closes without payment as a PRD recommendation; whether COD applies to services remains [OPEN DECISION].

9. Pre-order architecture

Pre-orders are an alternate action from the existing product detail page, not a second catalog. They apply only when a product/variant is Out of Stock and Admin has marked it pre-order-eligible.



Surface	Role	Required information / behavior
Product detail	Expose Pre-order instead of Add to Cart.	Selected product/variant and quantity; clear that availability date is not promised unless Admin enters one.
Pre-order request	Create request record.	Customer, product/variant, quantity, price at time of request, payment status, expected availability if entered, fulfillment status.
Admin queue	Review and Accept/Reject.	Acceptance enables payment; rejection is terminal; no stock reservation for a product not yet in stock.
Account detail	Make request visible to registered customer.	Review, payment, availability and fulfillment status; notifications on relevant changes.
Availability event	Join standard order fulfillment.	Confirmed paid pre-orders automatically enter standard fulfillment from Preparing onward when product becomes available.

[OPEN DECISION]
PRE-ORDER POLICY

The source does not settle whether the captured pre-order price remains honored if catalog price changes, nor allocation order among multiple confirmed pre-orders when stock becomes available.

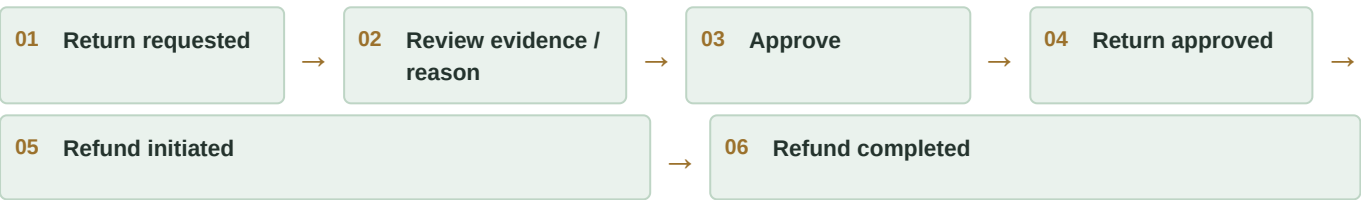
10. Returns, refunds and payment records

Customer return flow



Return access begins from a completed order. Admin may mark products non-returnable. The confirmed policy case is wrong item/damaged item; category-specific periods and rules remain unresolved.

Admin return and refund flow



Record	Meaning	Must remain distinct
Original Order Payment	Payment method and order payment state.	Separate from payment proof and refund.
Payment Proof	Customer screenshot(s) for non-COD payment; Admin review history.	Not a refund record; audit actor/action/timestamp.
Return Request	Customer's item-level request, reason and evidence.	Separate from original order state machine.
Refund Record	Manual refund completion with amount, method and timestamp.	Separate from original payment and payment proof; linked to return/order.

Return states

State	Customer view	Admin action
Return Requested	Request received.	Review.
Return Approved	Approval and refund processing visible.	Initiate/manual transfer and record.
Refund Initiated	Refund pending completion.	Record amount/method/timestamp when done.
Refund Completed	Refund record visible in return/payment history.	Closed.
Return Rejected	Decision and reason where captured.	Terminal.

[OPEN DECISION]
RETURNS POLICY

Category-specific return periods, detailed eligibility rules, non-returnable categories beyond Admin flags, and refund processing SLA are not finalized. Do not encode invented periods or promises.

11. Notifications and support

Notification architecture

Notifications originate from state changes in authoritative records. In-app notifications cover order, payment, service and pre-order lifecycle events. Email covers account verification, password reset and relevant transactional communication. WhatsApp is manually initiated, not an automated notification bus.

Event	Recipient	Channel / destination
Order submitted	Customer and Admin	In-app; email if available; real-time Admin/customer session where applicable.
Order accepted / rejected	Customer	In-app; reason where captured.
Payment proof submitted	Admin	In-app and real-time operational update.
Payment proof rejected / new proof requested	Customer	In-app; re-upload action.
Payment confirmed	Customer	In-app.
COD confirmation required	Customer	In-app action before Preparing.
Preparing / Ready for Pickup / Shipped / Out for Delivery / Delivered / Completed	Customer	In-app status timeline.
Service quotation sent / service status changes	Customer	In-app; Accept/Reject or progress context.
Pre-order availability	Confirmed pre-order customer	In-app; fulfillment begins when available.
Email verification / password reset	Customer	Email.
Return/refund update	Customer	In-app; refund record when completed.

WhatsApp placement

Context	Action	Boundary
Home / header / footer	Contact Al-Azhari Library on WhatsApp.	Persistent support access.
Product detail	Ask about product or availability.	Does not add to cart or create an order.
Cart / checkout	Get help with a question or edge case.	Checkout record remains authoritative.
Order detail / tracking	Contact library about this order.	Admin may use order context; no state change exists only in chat.
Services / service request	Clarify the request and exchange files.	Files exchanged via WhatsApp/Telegram; request record remains authoritative.
Admin order/customer/service detail	Contact Customer on WhatsApp.	Human communication channel; record decisions in Admin system.

REAL-TIME RULE

Socket.IO may push new orders, service requests, payment proof submissions and status changes, but the database remains the source of truth. If a live connection fails, the state must still persist and appear on reload.

12. Admin architecture and RBAC

Admin is a separate authenticated application shell, not a public navigation branch exposed to visitors. Its primary job is to move records through the approved lifecycles, maintain accurate catalog data, and keep the human review steps visible.

Admin sitemap

/admin

└─ Dashboard / operational queues

└─ Orders

└─ Pending Review

└─ Payment Verification

└─ Fulfillment

└─ Order Detail

└─ Catalog

└─ Products

└─ Product / Variant Editor

└─ Categories

└─ Customers

└─ Payments

└─ Services

└─ Service Catalog

└─ Requests

└─ Quotations

└─ Pre-orders

└─ Inventory

└─ Shipping

└─ Returns

└─ Refund Records

└─ Notifications

└─ Content / Homepage Merchandising

└─ Coupons & Discounts

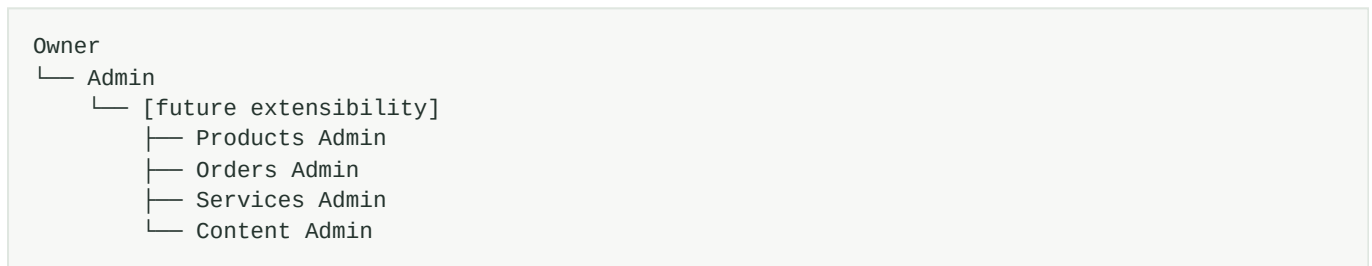
└─ Reports

└─ Settings / RBAC

Module	Pages	Workflow coverage	Permission boundary	Connections
Orders	Queue; detail	Review items/stock/customer info; Accept/Reject; status progression; cancellation window	Orders Admin future slice; Admin/Owner now	Order detail; payment; shipping; customer
Products	List; create/edit; variants	Name, category, price, images, description, availability; publish/unpublish	Admin/Owner	Inventory; catalog
Categories	List; edit	Maintain Books-first taxonomy; do not activate future categories in MVP	Admin/Owner	Products; merchandising
Customers	List; profile	View customer, orders, service requests, payment history; contact on WhatsApp	Admin/Owner	Orders; services
Payments	Verification queue; payment detail	Inspect screenshots; Confirm, Reject, Request New Proof; audit trail	Admin/Owner	Orders
Services	Catalog; request queue; detail	Manage confirmed services; review requests; create/send quotations; update lifecycle	Admin/Owner	Customer service requests
Pre-orders	Queue; detail	Accept/Reject; capture price and expected availability if entered; mark available	Admin/Owner	Products; fulfillment

Module	Pages	Workflow coverage	Permission boundary	Connections
Inventory	Stock list/detail	View/adjust stock; reserved vs available; final deduction at delivery/pickup	Admin/Owner	Products; orders
Shipping	Operational queue/settings	Select provider per order; review/adjust estimate; mark delivery states	Admin/Owner	Orders
Returns	Request queue/detail	Review item, reason, evidence; Approve/Reject	Admin/Owner	Refunds; orders
Refunds	Refund record	Record manual refund amount, method and timestamp; no automated gateway	Admin/Owner	Returns; payment history
Notifications	Operational list	Surface lifecycle events and real-time queue updates	Admin/Owner	All operational modules
Content	Product descriptions; homepage/seasonal merchandising	Edit descriptions; manage banners/featured products/categories without changing permanent taxonomy	Admin/Owner	Home; catalog
Coupons & Discounts	List; create/edit; activate/deactivate; usage	Percentage/fixed; seasonal; product/category-level; unresolved stacking/limits/expiry/threshold rules	Admin/Owner	Checkout; reports
Reports	Dashboards	Orders, revenue, outside-Qena geography, payment methods, service conversion, demand, pre-orders, coupon usage	Admin/Owner	Dashboard
Settings / RBAC	Settings; role boundary	Owner/Admin baseline; extensible permission model; future roles not initial UI	Owner/Admin as defined	All admin modules

RBAC architecture



Role	Initial status	Architecture boundary
Owner	Confirmed active role	Highest privilege; full administrative authority. Owner-only actions are not enumerated in the PRD, so do not invent a settings screen that implies them.
Admin	Confirmed active role	Day-to-day operations: orders, payment verification, inventory, shipping, services/quotes, returns/refunds, content and reports, subject to the configured permission model.
Future specialized roles	Extensible, not required in initial UI	Narrower slices of Admin capability; permission architecture should accept them later without redesign.

Unauthorized public/customer access should redirect to login where authentication is required. Authenticated users without permission should receive a clear Forbidden state and a safe route back. An Admin attempting an action reserved for Owner should be denied once such actions are explicitly defined.

Admin workflow anchors

Workflow	Operational sequence
Order review	Pending Review queue → order detail → verify items/stock/customer → Accept or Reject with reason where applicable.
Payment verification	Payment queue → inspect screenshot(s) → Confirm, Reject or Request New Proof → customer notification and audit record.

Workflow	Operational sequence
Inventory	Product/variant stock → view reserved vs available → adjust available stock; reservation on acceptance; final deduction on Delivered/Picked Up.
Service quotation	Service request → review/clarify → enter final price → send quotation → track customer Accept/Reject.
Shipping	Order → select provider → review/adjust estimate → advance Shipped/Out for Delivery/Delivered.
Return/refund	Return queue → review evidence → Approve/Reject → manually transfer refund → record amount, method, timestamp.
Content	Product descriptions and homepage/seasonal merchandising only; no taxonomy mutation as a side effect of a campaign.

13. Navigation system and routing strategy

Desktop storefront

Region	Contents	Behavior
Primary header	Logo; Shop; Services; Search; Account; Cart; WhatsApp.	Books-first Shop is the main discovery route; WhatsApp stays available without becoming the primary transaction path.
Catalog context	Breadcrumbs, category title, active filters, result count if available.	Return to parent category/search without losing the user's context.
Product context	Breadcrumbs; product identity; variant/availability; Add to Cart or Pre-order.	Keep support contextual and visible.
Checkout context	Step indicator; order summary; support action.	Do not mix service request flow into product checkout.
Footer	Contact/location/business information and support.	Final phone number remains [OPEN DECISION].

Mobile storefront

Mobile is a practical priority because existing customer behavior is concentrated in WhatsApp/Facebook. Keep the header compact: menu, logo, search and cart; make Account available from the menu/header; keep a persistent WhatsApp action. Use a single clear primary action on product and checkout screens, with visible availability, total and next step.

Customer dashboard

Use a compact sidebar on desktop and a stacked section/menu treatment on mobile. The active section should be explicit; breadcrumbs should show Account → section → detail. Current orders, service requests and notifications should be reachable without requiring a deep navigation path.

Admin shell

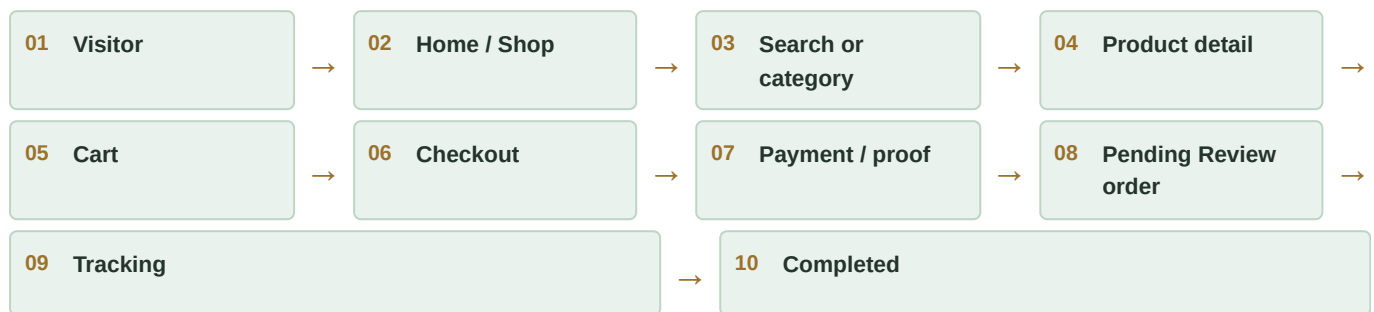
Use a persistent left sidebar grouped by Operations (Dashboard, Orders, Payments), Catalog (Products, Categories, Inventory), Services, Pre-orders, Shipping, Returns/Refunds, Customers, Content, Coupons, Reports and Settings/RBAC. A top bar can carry current role, notifications and contextual WhatsApp actions.

URL rules

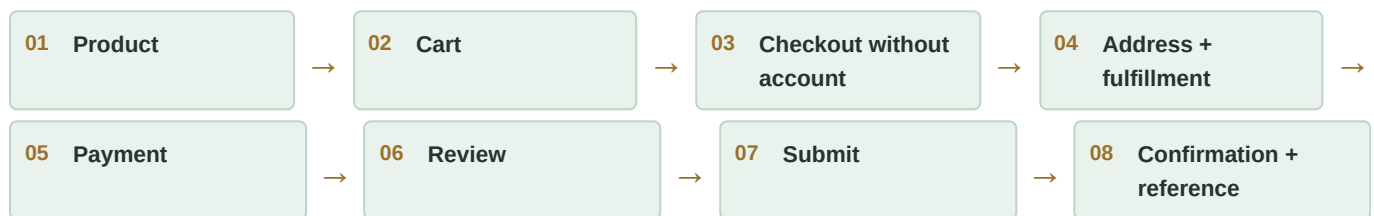
- Use clean, human-readable, SEO-supportive routes for public catalog and category pages.
- Use stable slugs for product/category/service pages; use IDs for private transactional records.
- Keep order, service request, pre-order, return and refund details under their owning domain rather than duplicating records in multiple navigation trees.
- Do not create a second catalog for pre-orders or a second checkout for services.
- Do not expose internal audit data or unauthorized payment-proof media in public routes.

14. User journeys and cross-page relationships

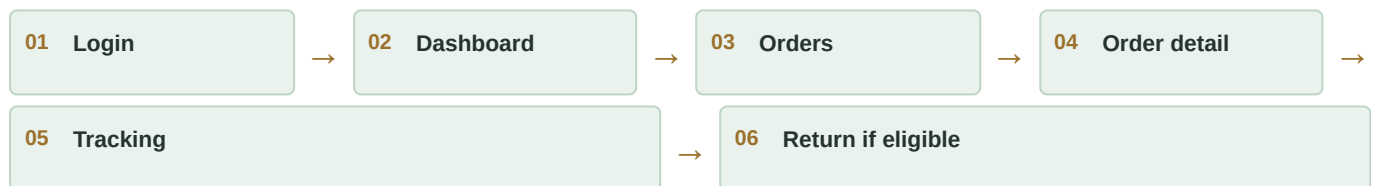
Journey A — Browse and purchase



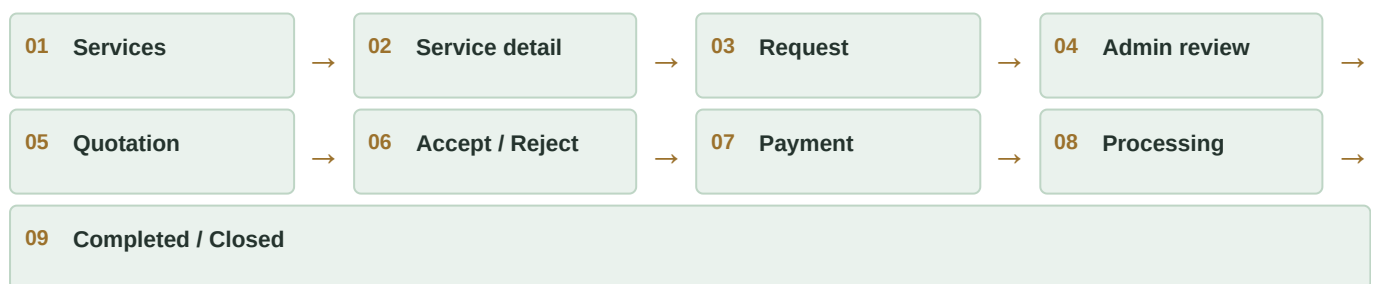
Journey B — Guest checkout



Journey C — Registered customer



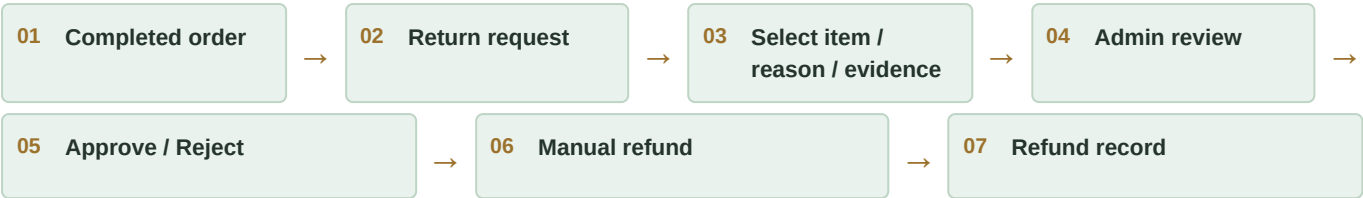
Journey D — Student service



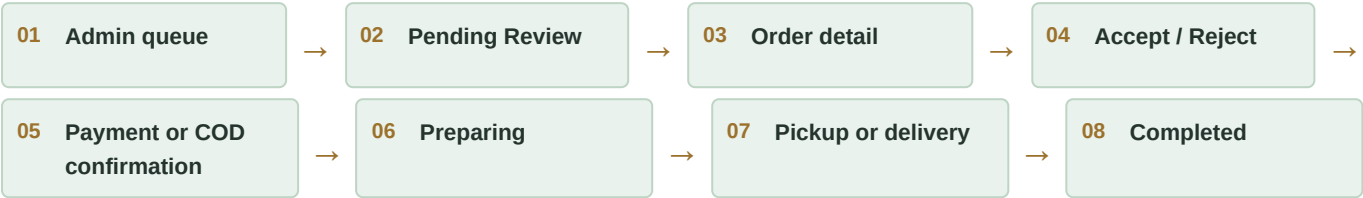
Journey E — Pre-order



Journey F — Return



Journey G — Admin order processing



Cross-page relationship matrix

From	To	Relationship	Trigger
Home	Shop / category	Discovery	Shop or category CTA
Product	Cart	Transaction	Add to Cart after applicable variant selection
Product	Pre-order	Alternate transaction	Pre-order CTA when out of stock + eligible
Cart	Checkout	Transaction	Checkout action
Checkout	Order	Creates authoritative record	Submit
Order	Tracking	Monitoring	View order / tracking
Order	Return	Post-purchase	Request Return from completed order
Services	Service Request	Separate service workflow	Request service
Service Request	Quotation	Admin review result	Admin sends final price
Quotation	Payment	Accepted service transaction	Customer Accepts
Account	Orders / Services	History and current state	Dashboard navigation
Admin	Order / Service / Payment	Operational control	Queue or notification action
Any relevant page	WhatsApp	Support	Contact action

15. Empty, error and edge states

The following states are architectural destinations or UI conditions required by the documented workflows. Copy, timeouts and policy details that are not specified must remain configurable or be marked open.

Area	State	Expected behavior
Cart	Empty cart	Explain there are no items; route to Shop/Categories; keep WhatsApp available.
Search	No results	Show query and active filters; allow clear/refine; return to catalog.
Product	Unavailable / not found	Do not offer purchase action when unavailable; not-found route must not expose internal data.
Product	Loading / API failure	Show recoverable loading/error state; retry and fallback to Shop/WhatsApp.
Checkout	Availability/price changed	Recommended re-validation point; inform customer before payment. Final behavior requires owner approval if treated as binding.
Checkout	Delivery not serviceable	Explain area is not currently serviceable; excluded-area list is open.
Checkout	Invalid/expired coupon	Reject with clear message and do not apply discount; expiry defaults/rules are open.
Payment	Missing proof	Prevent non-COD submission until screenshot is provided.
Payment	Proof rejected	Move to Awaiting New Proof; notify customer; allow re-upload.
Order	No orders	Account empty state with route to Shop.
Order	COD confirmation pending	Keep Customer Confirmation Required; no Preparing until confirmed; timeout is open.
Account	No addresses / no notifications / no payment history	Explain empty state and route to relevant action; do not fabricate records.
Services	No service requests	Route to Service Catalog; explain request-based lifecycle.
Services	No quotation yet	Do not expose payment action before Admin sends a quotation.
Returns	Non-returnable item	Do not offer request or show clear ineligible reason based on Admin flag.
Admin	Forbidden action	Deny action with clear permission message; preserve safe navigation.
System	Network/API failure	Do not imply a state change unless persisted; retry or refresh. Database remains source of truth.
System	Real-time connection unavailable	Persisted state remains visible on reload; live updates are not the data store.

Important unresolved edge rules

- COD non-confirmation timeout/expiry is not defined.
- Partial acceptance when only part of a multi-item order has stock is not defined.
- Concurrent last-unit contention is not defined.
- Pre-order allocation order and price-honoring policy are not defined.
- Final shipping-cost binding point and re-confirmation behavior are not defined.
- Return periods, detailed eligibility and refund SLA are not defined.
- Coupon stacking, usage limits, expiry defaults, minimum order value and customer restrictions are not defined.
- Accessibility conformance, performance targets and availability SLAs are not defined.

16. IA validation summary

Coverage

This IA gives every major PRD module a home: public storefront, permanent catalog, search/filtering, product detail and variants, cart, checkout, six-method PRD payment list, proof-of-payment review, delivery/pickup, order lifecycle/tracking, authentication, customer dashboard, services/quotations, pre-orders, returns/refunds, notifications, contextual WhatsApp, Admin modules, reports, content, coupons and RBAC. The seven major journeys have explicit entry and exit paths.

Potential gaps

No additional product gap was invented beyond the source material. The main implementation gaps are unresolved source decisions: service-specific forms and turnaround times, shipping-rate logic and policy, return rules, final payment-method list, role-specific permissions, and operational timeouts/edge handling. These are listed as open rather than hidden in page definitions.

Open decisions register

ID	Decision
OD-01	Final phone numbers for website and WhatsApp actions.
OD-02	Email-verification gating scope.
OD-03–06	Shipping rate source/logic, final-cost binding, delivery ranges and excluded areas.
OD-07	COD non-confirmation timeout/expiry.
OD-08–09	Concurrent stock contention and low-stock/reorder thresholds.
OD-10–11	Pre-order price honoring and allocation order.
OD-12–14	Service fields/documents, turnaround times and whether COD applies to services.
OD-15–16	Category return policies/periods and refund processing SLA.
OD-17	Coupon stacking, limits, expiry, minimum order value and customer restrictions.
OD-18–19	Numeric KPI targets; accessibility, performance and availability targets.
OD-20	Final payment-method list: PRD six-method list versus Brief's four-method list including cash at library.
OD-21	Partial-acceptance behavior for multi-item orders with insufficient stock.

Architecture notes to preserve

- Books remain the primary storefront focus; Azhar specialization does not exclude general-education or university customers.
- Services remain structurally separate from product commerce and never show a final fixed price before Admin quotation.
- Seasonal merchandising changes presentation, not permanent taxonomy.
- The website is the source of truth; WhatsApp is human support and communication.
- Guest checkout remains available.
- Non-COD payment proof is reviewed manually; there is no automated payment gateway and no transaction-ID field.
- Pickup is free; the library selects the delivery provider.
- Product orders and service requests are separate entities with separate lifecycles.
- Admin is isolated behind RBAC and future specialized roles must fit without redesign.
- No unsupported feature or business rule should be inferred from a route placeholder.

Final implementation handoff

A UX designer can begin wireframes from the page inventory and state tables; a frontend developer can implement the public, account and Admin route shells; a backend developer can preserve the product-order-service-payment-inventory boundaries; and a product owner can review the open-decision register before these items become hard-coded behavior.

SOURCE NOTE

This document is an architecture translation, not a replacement for the two source documents. Where the PRD and Master Product Brief differ, the PRD's explicit functional decision is followed and the discrepancy is surfaced.